

Math 151A - Homework 2
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1. Newton's Method and Local Convergence

For $f(x) = \cos x + \sin^2(50x)$,

$$f'(x) = -\sin x + 100 \sin(50x) \cos(50x), \quad f''(x) = -\cos x + 10000 \cos(100x) - \sin x.$$

Near $p = \pi/2$, we estimate $|f'(x)| \approx 49$ and $|f''(x)| \leq 10002$. By Theorem 2.6,

$$|x_0 - p| < \frac{2|f'(p)|}{|f''(\xi)|} \approx 0.01.$$

Thus Newton's method converges if $|x_0 - \pi/2| < 0.01$.

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2. Equivalent Secant Forms

The usual update is

$$p_{n+1} = p_n - f(p_n) \frac{p_n - p_{n-1}}{f(p_n) - f(p_{n-1})}.$$

An equivalent form is

$$p_n = \frac{f(p_{n-1})p_{n-2} - f(p_{n-2})p_{n-1}}{f(p_{n-1}) - f(p_{n-2})}.$$

When $p_{n-1} \approx p_{n-2}$, the second form is more stable. If $f(p_{n-1}) \approx f(p_{n-2})$, use the standard form instead. A simple hybrid rule is:

If $|p_{n-1} - p_{n-2}| < 10^{-8}$, use the fraction form; otherwise use the standard form.

Choose the version that avoids small denominator differences.

3. Newton's Method for $f(x) = x^2 - 1$

For $f(x) = x^2 - 1$, $f'(x) = 2x$,

$$x_{n+1} = x_n - \frac{x_n^2 - 1}{2x_n} = \frac{1}{2} \left(x_n + \frac{1}{x_n} \right).$$

Starting from $x_0 = 100,000$:

$$x_1 = 50,000.000005, \quad x_2 = 25,000.000010, \quad x_3 = 12,500.000020, \dots$$

The iterates halve each step until they approach 1.

For large x_n ,

$$x_{n+1} \approx \frac{x_n}{2} \Rightarrow r_n = \frac{|x_{n+1} - 1|}{|x_n - 1|} \approx \frac{1}{2}.$$

Thus convergence is initially linear with ratio 1/2, and becomes quadratic near the root.

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{1}{x_n} \right), \quad x_\star = 1.$$